

Chan Nhu (Soleil) Nguyen Pham

Undergraduate Researcher, Department of Computer Science, Purdue University

soleilpham@gmail.com • (765) 806-0753 • github.com/soleilpham • soleilpham.github.io/source • LinkedIn

RESEARCH INTERESTS

Robot manipulation with statistical safety guarantees; wearable and ubiquitous sensing for health; and human-centered security and privacy in AI-enabled extended reality (XR). I build reliable, well-tested research systems and ship production software used by real organizations.

EDUCATION

Purdue University

West Lafayette, IN

B.S. in Computer Science, Minor in Mathematics

Aug 2024 – May 2027 (expected)

- **Concentrations:** Machine Intelligence; Systems Software.
- **Relevant Coursework:** Data Mining & Machine Learning, Artificial Intelligence, Data Structures & Algorithm Analysis, Systems Programming (C, Assembly), Computer Architecture, Compilers, Operating Systems, Object-Oriented Programming (Java), Competitive Programming (C++), Linear Algebra, Multivariable Calculus, Applied Regression.

Saigon South International School

Ho Chi Minh City, Vietnam

High School Diploma

Graduated May 2024

PUBLICATIONS & MANUSCRIPTS

*Equal contribution.

- [1] F. Ye*, **S. Pham***, A. Durbha, R. Li, and Q. Xue. “EarringBite: Automatic Eating Detection and Chewing Analysis with Smart Earrings.” *Under review at ACM CHI Conference on Human Factors in Computing Systems (CHI)*, 2027. [Code]

RESEARCH EXPERIENCE

EarringBite: Automatic Eating Detection with Smart Earrings

Purdue University

Co-First Author — under review at ACM CHI 2027

May 2026 – Sep 2026

- Architected a 3-stage pipeline that synchronizes bilateral smart-earring clocks and aligns 21.6M rows of accelerometer and PPG data across 12 participants using timestamp-based windowing.
- Developed a multimodal eating-detection system with dual ExtraTrees classifiers over 450 engineered features, achieving **96.45% accuracy**, **0.9638 macro-F1**, and 95.8% daily-wear specificity.
- Outperformed 149 deep-learning configurations with the feature-based model, and released the code and a Flask web demo.

INDUSTRY EXPERIENCE

Purdue University — College of Liberal Arts

West Lafayette, IN

Web Developer

Jan 2026 – Present

- Maintain and extend web services for the entire College of Liberal Arts, building reusable templates and components (HTML, CSS, JavaScript, Apache Velocity).
- Resolve support tickets spanning content updates, WCAG accessibility improvements, and legacy-system modernization.

Reclub — Sports Technology Startup

Ho Chi Minh City, Vietnam

Software Engineer Intern

May 2025 – Aug 2025

- Shipped an automated video-segmentation system that detects pickleball rallies in raw match footage, cutting per-match editing time from hours to about 5 minutes and enabling automated highlight generation at platform scale.
- Built a motion-aware YOLOv10s classification pipeline that achieved 96% accuracy and was optimized for near-real-time, on-device inference.
- Owned the ML lifecycle end to end, from labeling 36K+ frames to training and deployment on Docker, Google Cloud Storage, and Vertex AI, taking the system to production inference on live user content.

Amazon Web Services (AWS) Youth Experience Program

Data Analytics Intern

Remote

Jul 2023 – Aug 2023

- Built Python pipelines (Pandas, NumPy, scikit-learn) to predict forest type for wildfire-risk analysis, comparing logistic regression, decision trees, random forests, neural networks, and KNN.
- Recognized as a top-5 intern project for technical quality and clarity of analysis.

LEADERSHIP & SERVICE

Purdue ACM SIGAPP — Student App Development Club

Team Lead (2026–27) • President (2025–26) • Member (2024–25)

West Lafayette, IN

Aug 2024 – Present

- **President (2025–26):** Led a 30-member club running 3 concurrent product teams (BoilerBuzz, BoilerRooms, and the PGSG Grant Portal), owning architecture decisions, sprint planning, code-review standards, and onboarding.
- Secured and managed the club's first external client, Purdue Graduate Student Government, and continue to lead the portal's integration with Purdue systems.
- Ran the club's 3-track workshop series: **frontend** (React and React Native in JavaScript/TypeScript), **backend** (C#/.NET and Express.js), and **dev tools** (Git, Docker, and CI/CD).
- Delivered 7 workshops and 8 events in total in 2025–26: all 3 tracks for club members; backend (Express.js) and frontend (React/React Native with UI/UX) workshops for the Purdue Computer Science Women's Network (CSWN); frontend workshops for Girls Who Code and the Innovate Her hackathon; and a Capital One networking event co-hosted with CSWN.
- **Team Lead (2026–27):** Delivered all 3 workshop tracks to club members so far this year, expanded the CSWN partnership to recurring workshops and career events, and launched a collaboration with AIGA Purdue, the student chapter of the professional design association, to connect engineers with graphic and UI/UX designers.
- Standardized engineering across projects with one-command dev scripts, Docker Compose stacks with seeded databases, CI/CD pipelines, and onboarding guides.
- Led the club website rebuild (Vue 3, Vite, Tailwind CSS), authoring 26 of 39 commits.

Unity Impact — Game Development Club, Saigon South International School

Co-Founder & President

Ho Chi Minh City, Vietnam

Aug 2022 – May 2024

- Co-founded and led a student game-development club whose members published 8 browser games on itch.io.
- Built and published 2 of the games in Unity and C#: *Save the Chicken*, a reaction-time measurement simulation, and *Airplane Shooter*.

SOFTWARE PROJECTS

PGSG Grant Portal | FastAPI, PostgreSQL, React 19, MUI, Keycloak, Docker, Jenkins, Playwright

Nov 2025 – Present

- Grant application, review, and decision platform for Purdue Graduate Student Government covering 12 grant programs, including travel, research, professional development, and childcare (21 contributors, 343 commits, 62 endpoints, ~17K lines of code).
- Set up the base project, converted all 12 grant specifications into JSON form templates with conditional-visibility expression trees, and built phase-1 features for all 3 user roles.
- Implemented Keycloak OIDC with role-based access control, and built double-blind 3-reviewer scoring across 10 criteria, review seasons, batch decisions, decision release, and reimbursement.
- Shipped Dockerized services with Jenkins CI/CD, database migrations, and Playwright end-to-end security and lifecycle tests on a self-hosted server.
- Leading final testing, review, and integration with Purdue SSO and university databases in partnership with PGSG ahead of campus deployment.

BoilerBuzz | React Native (Expo), TypeScript, Express 5, PostgreSQL, Clerk

Aug 2025 – Mar 2026

- Campus event and poster feed with search, pinning, bookmarks, and club management; led a 21-contributor team across frontend and backend.
- Built PostgreSQL connection pooling, navigation, theming, and feed and search components, plus the typed API client connecting ~30 REST endpoints to 6 screens.

BoilerRooms | React Native (Expo), ASP.NET Core (C#), MongoDB, Google Maps API

Sep 2024 – Present

- Campus buildings and housing app with map outlines, search, reviews, and events, backed by a 31-endpoint REST API.
- **As a member:** Built the events CRUD API, a Google Maps distance endpoint, a building-outline (polygon)

service, and polymorphic housing, room, and dining models with filter endpoints.

- **As team lead:** Refactored 110+ files, removing broken Auth0 and video-tour code; added a Google Maps service, Docker Compose with a seeded MongoDB, a frontend API layer with shared hooks and session context, and a theme system; and planned a 28-issue feature backlog.

Site Scouter — Catapult 2026 | *FastAPI, PyTorch, React, Leaflet, OpenStreetMap, Docker, AWS* Mar 2026

- Geospatial tool that divides a user-drawn region into cells and scores each for solar, wind, and data-center suitability from satellite imagery, terrain, and OpenStreetMap data; its PyTorch model, trained on thousands of real projects, predicts power generation with $R^2 > 0.90$.

Talking Slides — BoilerMake 2025 | *Django REST, MongoDB/GridFS, Redis, React, Auth0, OpenAI API* Feb 2025

- Converts PDF study materials into multi-voice, podcast-style audio using PyMuPDF text extraction, LLM script generation, and speech synthesis, with Redis caching for fast playback.

BoilerGram — CS 180 Team Project | *Java, TCP Sockets, Multithreading, Swing* Fall 2024

- Instagram-style social platform built without a web framework on a multithreaded client-server architecture over raw TCP, with authentication, following and blocking, file-based persistence, and real-time group messaging (196 commits).

Unity Games — Hexagon World and Save the Chicken | *C#, Unity, WebGL* 2022 – 2025

- Procedural hexagonal-grid terrain generation and a reaction-time game, both exported to WebGL and hosted on GitHub Pages.

HONORS & HACKATHONS

- **3rd Place**, Hello World Hackathon, Purdue University (2024) — *Roadie Rescue*, a MERN and Leaflet app that connects stranded drivers with nearby volunteers.
- **Top-5 Intern Project**, AWS Youth Experience Program (2023).
- **Participant**, Catapult (2026) — *Site Scouter*; BoilerMake (2025) — *Talking Slides*.

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, Java, C++, C, C#, SQL, MATLAB, Assembly.
- **Software Engineering:** FastAPI, Node.js/Express, ASP.NET Core, Django, React, React Native/Expo, Vue, PostgreSQL, MongoDB, Redis, Keycloak/OIDC, REST, Docker, Jenkins, GitHub Actions, Playwright, GCP (Vertex AI, GCS), AWS.
- **ML & Robotics:** PyTorch, scikit-learn, NumPy, Pandas, Numba, MuJoCo, YOLO, conformal prediction, reinforcement learning (PPO), LLM planning.
- **Signals & Graphics:** PPG/IMU signal processing, SPH fluid simulation, three.js, Unity, FFmpeg.